Development & Field Testing of a Technology-Based Inferencing Intervention

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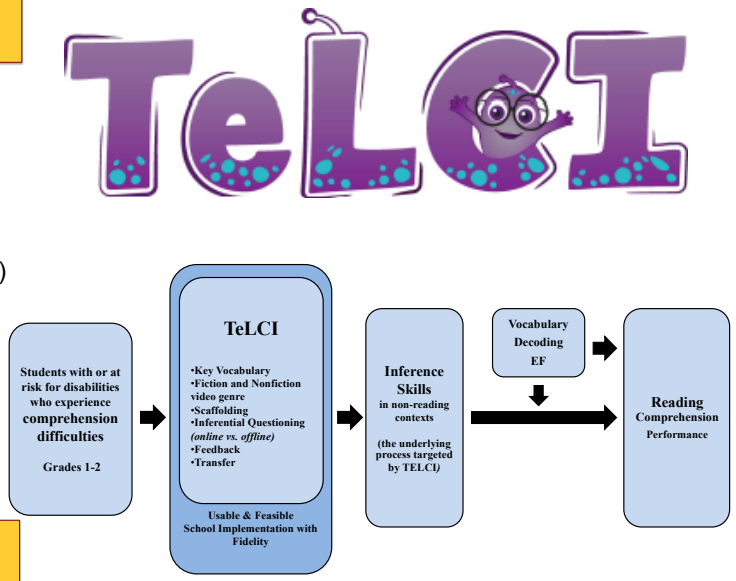
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Purpose of TeLCI

Designed to improve reading comprehension by developing **inference making** for **struggling readers** in **Grades 1-2**. TeLCI does not rely on decoding skills. It is an interactive, cloud-based software application with 24 **modules**, which engage students to:

- View age-appropriate ***videos*** (12 *fiction*, 12 *nonfiction*)
- Learn key academic ***vocabulary*** words
- Respond (by touch screen selections) to ***inferential questions***
- Receive ***scaffolding*** and ***feedback*** for each question
- Receive ***small-group*** transfer lessons using book-reading



Theoretical Background

Inference making is one of the unique, significant *predictors of reading comprehension,* with some studies indicating a causal link from poor inference making to poor reading comprehension (Oakhill & Cain, 2012).

The ability to draw inferences is a *general skill*—it is not specific to reading (Kendeou et al, 2008; Kendeou, 2015).

Inference skills can be improved using **questioning that includes scaffolding and specific feedback** (McMaster et al., 2012)

Technology provides a cost-effective, standardized, individualized delivery of instructional tools in classroom settings that **can improve various learning outcomes** (McNamara et al., 2017)

Usability and Feasibility

Participants

- *N* = 95 students in Grades 1 and 2

52% female, 54% White, 19% Black, 21% Hispanic

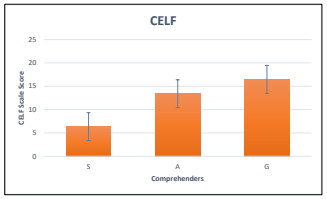
47% reduced/free lunch, 23% ELL

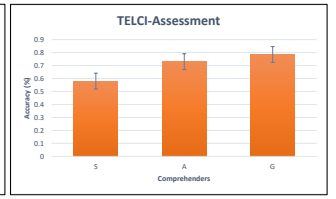
- *N* = 8 Teachers; *N* = 2 Literacy Experts

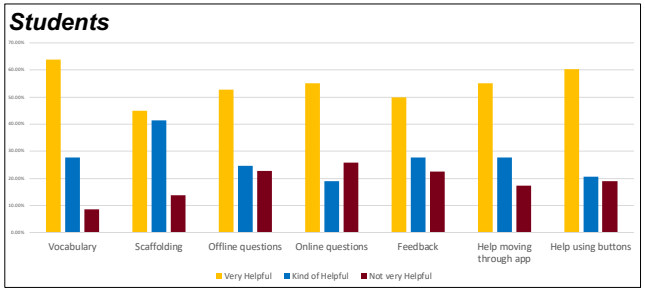
Methods

- Students' Scores (CELF, QRI, MEFS, TeLCI Assessment, TeLCI module data)
- Usability and Feasibility Observations and Questionnaires
- Focus Groups (Teachers, Literacy Experts)

Selected Results







Teachers

- varied in their opinion ***of who will benefit from TeLCI***
- rated most TeLCI features as ***appropriate for children in Grades 1-2***
- provided ***valuable input*** on how to improve TeLCI features (e.g., feedback and motivation)
- envisioned using TeLCI at least ***2-3 times per week*** as part of ***guided independent work time***

Next Steps

Field Test: Online vs. Offline Questioning (Year 2)
Pilot Test: Efficacy RCT Study (Year 3)

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